



UNIVERSITY OF THE PELOPONNESE & NCSR “DEMOCRITOS”
MSC PROGRAMME IN DATA SCIENCE

Monitoring the quality of Internet of Things services

Subtitle

by

Stavroula Pouli

A thesis submitted in partial fulfillment
of the requirements for the MSc
in Data Science

Supervisor: Charilaos Akasiadis
Post-Doctoral Researcher

Co-supervisors: Co-Supervisor name, Co-Supervisor2 name
Co-Supervisor title, Co-Supervisor2 title

Athens, September 2026

Monitoring the quality of Internet of Things services

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MSc. Thesis, MSc. Programme in Data Science

University of the Peloponnese & NCSR “Democritos”, September 2026

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Athens, September 2026



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Stavroula Pouli

Athens, September 2026

Acknowledgments

Acknowledgment text ...

To my family.

Abstract

The Internet of Things (IoT) refers to a network of physical objects embedded with sensors, software, and other technologies that connect and exchange data with other devices and systems over the internet. The growing complexity of IoT calls for reliable methods to monitor the quality and performance of interconnected devices. This thesis investigates the use of the Matrix Profile, a simple yet powerful technique for time-series analysis, to detect anomalies that indicate service degradation in IoT systems. The proposed approach is evaluated on univariate and multivariate datasets and compared against established statistical and AI-based algorithms to assess its effectiveness, interpretability, and computational efficiency. [Note: This section will later include the main findings and comparative results once the evaluation is completed.]

Keywords: Anomaly detection, Internet of Things, Matrix Profile, Time-Series Analysis

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List of Abbreviations

B-PCA	Bilateral Principal Component Analysis
CNN	Convolutional Neural Network
EA	Evolutionary Algorithm
IMP	Interval Matrix Profile
IoT	Internet of Things
LAN	Local Area Networks
LOF	Local Outlier Factor
MP	Matrix Profile
MTS	Multivariate Time Series
MTSAD	Multivariate Time Series Anomaly Detection
ODIN	Outlier Detection using In-degree Number
RDOS	Relative Density-based Outlier Score
ROGE	Robust Genetic Algorithm
STL	Signal Temporal Logic
STREL	Spatio-Temporal Reach and Escape Logic
UTS	Univariate Time Series
WAN	Wide Area Networks

Chapter 1

Introduction

1.1 Introduction

The Internet of Things (IoT) connects millions of sensors and smart devices that communicate and share information without human intervention. These systems continuously generate large volumes of time-series data that need to be processed in near real-time. Analyzing such data efficiently requires methods that can work under limited computational resources while maintaining accuracy and reliability.

In IoT environments, two major challenges often arise: *concept drift* and *anomaly detection*. Concept drift occurs when the statistical behavior of the data changes over time, making previously trained models less accurate [1]. Anomaly detection (also called outlier or deviation detection) refers to the identification of unusual patterns that deviate from expected system behavior [2]. The need to recognize unexpected behaviors has led to numerous algorithms, ranging from classical statistical approaches to modern machine-learning and artificial-intelligence techniques [3]. However, many of these methods require extensive training data, heavy computation, or complex parameter tuning, which limits their applicability in resource-constrained IoT systems.

Recently, the Matrix Profile (MP) framework has emerged as a powerful data-driven approach for time-series analysis. First introduced by Yeh *et al.* [4], MP has attracted considerable attention due to its versatility, simplicity, and scalabil-

ity. Unlike model-based approaches, MP relies on distance computation between subsequences, allowing efficient discovery of motifs (repeated patterns) and discords (anomalies) without prior assumptions or training data.

1.2 Problem description

The rapid growth of Internet of Things (IoT) technologies has created an urgent need for anomaly-detection methods that are both reliable and computationally efficient. Traditional supervised and model-based techniques often struggle to perform well in dynamic environments, where data patterns can change over time and labeled examples are limited. These challenges can lead to delayed or inaccurate detection of abnormal behavior, affecting the overall reliability and safety of IoT systems. This study addresses the need for lightweight and explainable approaches that can identify anomalies directly from raw IoT time-series data, without relying on frequent model retraining or complex feature engineering. In particular, it investigates the potential of the Matrix Profile to detect both individual (point-wise) and recurring (pattern-based) anomalies in temperature sensor data, while maintaining a balance between accuracy, interpretability, and computational efficiency.

1.3 Thesis Contribution

This work contributes to the field of IoT anomaly detection by exploring the Matrix Profile (MP) as a practical and explainable technique for analyzing sensor-based time-series data. It demonstrates how MP can be effectively applied to temperature-based IoT datasets in both univariate and multivariate settings, highlighting its strengths in detecting subtle and recurring anomalies without the need for complex model training. Additionally, the study proposes a reproducible experimental framework that supports scalable monitoring applications in sensor-driven environments.

1.4 Thesis structure

The remainder of the present study is organized as follows. Section 2 reviews the theoretical background and related work on IoT anomaly detection. Section

3 describes the research methodology adopted in this study. Section 4 provides a detailed description of the datasets employed and the preprocessing steps applied. Section 5 reports and analyzes the experimental evaluation results, comparing Matrix Profile performance with traditional methods. Finally, Section 6 presents the conclusions and outlines possible directions for future research.

Chapter 2

Background & Related Work

This section provides an overview of the fundamental concepts that underlie anomaly detection in IoT. It begins by defining the key characteristics of the time-series data, followed by an explanation of concept drift and the different types of anomalies. The section then reviews related research, summarizing major algorithms, methodologies, and recent developments in the field, with particular emphasis on approaches relevant to time-series and IoT data analysis.

2.1 Background

2.1.1 Time-series

Time-series data consist of sequences of observations collected at uniform time intervals, capturing how a given phenomenon evolves over time. This temporal structure allows the analysis of dynamic behaviors and helps identify how values change in chronological order. Each observation x_t represents the state of a system at time t , and examining the full sequence enables the discovery of patterns such as long-term trends, seasonality, cycles, and irregular fluctuations [5]. As noted in [6], time series are often divided into univariate (one-dimensional) and multivariate (multi-dimensional) forms, which are described below.

2.1.1.1 Univariate Time Series

A univariate time series (UTS) is a sequence of data points corresponding to a single variable that changes over time. It can be expressed as:

$$X = (x_1, x_2, \dots, x_t)$$

where x_i represents the data value at timestamp $i \in T$, and $T = \{1, 2, \dots, t\}$.

2.1.1.2 Multivariate Time Series

A multivariate time series (MTS) represents multiple variables that evolve together over time, where each variable is influenced by both its past values (temporal dependency) and by correlations with other variables (spatial or inter-metric dependency). An MTS can be represented as a sequence of vectors over time:

$$X = (X^1, X^2, \dots, X_t) = \{(x_1^1, x_1^2, \dots, x_1^d), (x_2^1, x_2^2, \dots, x_2^d), \dots, (x_t^1, x_t^2, \dots, x_t^d)\}$$

where $X_i = (x_i^1, x_i^2, \dots, x_i^d)$ represents a data vector at time i , each x_i^j indicates the observation at time i for the j^{th} dimension, and $j = 1, 2, \dots, d$, where d is the total number of dimensions.

2.1.2 Concept Drift

Concept drift is an evolution of data that invalidates an existing model when the underlying data distribution changes over time [7]. In the context of classification learning [8], let $P(Y)$ denote the prior probability distribution over the class labels, $P(X)$ the prior probability distribution over independent variables (covariates), and $P(X, Y)$ the joint probability distribution over objects and class labels. The conditional probability $P(Y | X)$ represents the likelihood of observing a class label given the input features, whereas $P(X | Y)$ describes the posterior distribution of independent variables conditioned on class labels.

To capture temporal dynamics, let $P_t(X)$ denote the probability distribution

at time t . Given a joint distribution (concept) $P(X, Y)$ where Y is the target variable (if applicable), concept drift occurs between times t and $u = t + \Delta$ when the distributions change:

$$P_t(X, Y) \neq P_u(X, Y)$$

where Δ represents the difference in time (time interval) between the two observations t and u .

Concept drift can be categorized into several types [7]. *Real drift* occurs when the target concept itself changes, altering the relationship between inputs and outputs. *Virtual drift* involves changes in the distribution of input features without affecting the underlying concept. *Class prior drift* refers to variations in the proportion of class labels over time. When such changes occur suddenly, the phenomenon is known as *abrupt drift*; when they evolve gradually, it is called *gradual drift*. Finally, *recurrent drift* describes cases in which previously observed concepts reappear periodically, often due to seasonal or cyclic patterns.

2.1.3 Anomalies

Anomalies are occurrences in a dataset that are unusual and do not conform to a well-defined notion of normal behavior [2, 9]. In the Internet of Things (IoT) domain, they refer to deviations from expected behavior within devices, networks, or sensor data streams. Detecting such anomalies is crucial for maintaining the integrity, reliability, and security of IoT services. As the number of connected devices increases, identifying abnormal patterns quickly and accurately becomes essential for ensuring service quality and preventing potential faults or attacks.

Anomalies are commonly categorized into three main types: *global (or point)*, *contextual*, and *collective* [10]. *Global* or *point anomalies* refer to individual data points that significantly deviate from the rest of the dataset. In IoT networks and wireless sensor systems, such outliers can generally be classified as *errors*, *events*, or *malicious attacks* [11]. Errors are typically caused by faulty sensor readings or random noise and, although common, they reduce the reliability of collected data.

Events, on the other hand, represent meaningful environmental changes such as temperature spikes, air-quality variations, or natural phenomena, which are often sustained over time. Malicious attacks occur when compromised sensors intentionally transmit false data, posing serious security risks to the entire network.

Contextual anomalies occur when the anomaly is not defined solely by the individual data point but depends on the context or relationship among data points. *Collective anomalies* involve anomalous patterns or behaviors that can only be identified by considering a group or collection of data points together. In multivariate time-series data, relationships among sensors and variables are complex [12], making it less precise to classify anomalies as purely point, contextual, or collective. Even if individual sensor readings appear normal, a combined analysis may reveal patterns inconsistent with typical inter-sensor dependencies.

2.1.4 Matrix Profile

The *Matrix Profile (MP)* is a data structure that enables the efficient discovery of patterns and anomalies in time-series data. It provides, for each subsequence in a time series, the distance to its most similar subsequence, allowing the identification of both *motifs* (repeated patterns) and *discords* (unusual subsequences) [4, 13]. Let a time series $T \in \mathbb{R}^n$ be a sequence of real-valued observations: $T = [t_1, t_2, \dots, t_n]$, where n is the length of the series.

2.1.4.1 Definition 1

A subsequence $T_{i,m} \in \mathbb{R}^m$ is defined as a continuous subset of the original time series T , starting at position i and containing m consecutive observations. Formally:

$$T_{i,m} = [t_i, t_{i+1}, \dots, t_{i+m-1}], \quad i = 1, 2, \dots, n - m + 1$$

The length of a subsequence cannot exceed that of T ($1 \leq m \leq n$), and its starting index must satisfy $1 \leq i \leq n - m + 1$ to ensure that it remains entirely within the bounds of T [14].

2.1.4.2 Definition 2

A *time-series motif* is defined as the pair of subsequences that exhibit the highest similarity (i.e., the smallest distance) among all possible subsequence pairs within the series. Formally, two subsequences $T_{a,m}$ and $T_{b,m}$ form a motif pair if

$$\text{dist}(T_{a,m}, T_{b,m}) \leq \text{dist}(T_{i,m}, T_{j,m}), \quad \forall i, j \in [1, 2, \dots, n - m + 1], a \neq b, i \neq j$$

where dist denotes the z -normalized Euclidean distance between two subsequences.

To ensure that similarity comparisons are independent of scale and offset, each time series T is first transformed into its z -normalized form. The normalized series $\hat{T}$ is obtained by adjusting T to have mean $\mu = 0$ and standard deviation $\sigma = 1$. For a subsequence $T_{i,m} = [t_i, t_{i+1}, \dots, t_{i+m-1}]$, the normalized version is computed as:

$$\hat{T}_{i,m} = \frac{T_{i,m} - \mu_i}{\sigma_i}$$

where μ_i and σ_i are the mean and standard deviation of the subsequence $T_{i,m}$, defined as:

$$\mu_i = \frac{1}{m} \sum_{k=0}^{m-1} t_{i+k}, \quad \sigma_i = \sqrt{\frac{1}{m} \sum_{k=0}^{m-1} (t_{i+k} - \mu_i)^2}$$

2.1.4.3 Definition 3

The distance between two z -normalized subsequences $\hat{T}_{i,m}$ and $\hat{T}_{j,m}$ is computed using the Euclidean distance [15]:

$$\text{dist}(T_{i,m}, T_{j,m}) = \sqrt{\sum_{k=1}^m (\hat{t}_{i+k-1} - \hat{t}_{j+k-1})^2}$$

For a given subsequence $T_{i,m}$, the *distance profile* $\mathbf{D} \in \mathbb{R}^{n-m+1}$ represents the

vector of distances between $T_{i,m}$ and all other subsequences in T :

$$D[j] = \text{dist}(T_{i,m}, T_{j,m}), \quad \forall j \in [1, 2, \dots, n - m + 1], i \neq j$$

2.1.4.4 Definition 4

The *Matrix Profile* $\mathbf{P} \in \mathbb{R}^{n-m+1}$ is then defined as the collection of the minimum distances for each subsequence:

$$P[i] = \min_{i \neq j} \text{dist}(T_{i,m}, T_{j,m})$$

Thus, the lowest values in P correspond to *motifs*, representing highly similar repeating patterns, while the highest values indicate *discords*, or anomalous subsequences that deviate significantly from the rest of the series.

To prevent *trivial matches*—cases where a subsequence is compared with itself or another that largely overlaps in position—the Matrix Profile employs an *exclusion zone*. This exclusion region, typically set heuristically to half the subsequence length ($m/2$), ensures that identified motifs correspond to truly distinct and non-overlapping patterns within the time series [13].

The concepts introduced in this section provide the theoretical foundation for understanding the anomaly detection techniques presented in the following discussion.

2.2 Related Work

Once the importance of anomaly detection in IoT services has been established, the next critical step is to examine the existing landscape of techniques and algorithms used for this purpose. Researchers have proposed a wide variety of methods, ranging from formal-logic monitoring to machine-learning and deep-learning approaches.

A recent comprehensive survey by Fährmann *et al.* [16] emphasizes the importance of anomaly detection for ensuring the safety and reliability of smart envi-

ronments. Their study provides an extensive overview of state-of-the-art anomaly detection techniques applied to multivariate time-series data in domains such as smart homes, smart transport, and smart industry. The authors review a wide range of methods—spanning statistical, proximity-based, and deep-learning approaches—and also discuss key aspects such as anomaly categorization, detection scenarios, evaluation metrics, and available benchmark datasets. This work highlights the increasing research focus on adaptive and explainable methods that can effectively handle large-scale, dynamic IoT data streams.

Beyond these general frameworks, several research efforts have explored specific algorithms and models for monitoring and anomaly detection in IoT systems. In [17], the authors present an offline monitoring algorithm for *Spatio-Temporal Reach and Escape Logic (STREL)* and demonstrate its application in monitoring a simulated mobile ad-hoc sensor network. Their future work focuses on distributed monitoring algorithms to extend STREL for large-scale cyber-physical systems and on studying its expressive power for spatio-temporal reasoning. In [18], a framework is introduced for learning *Signal Temporal Logic (STL)* specifications from labeled datasets of normal and anomalous trajectories. It employs a robust genetic algorithm (ROGE) that combines an *Evolutionary Algorithm (EA)* for formula-structure learning with a Gaussian-process-based optimization for parameter synthesis. The algorithm appears to be specialized for certain types of anomaly detection and classification tasks.

Machine learning has been widely used to detect outliers and abnormal patterns in IoT data. The study in [11] tackles the critical challenge of outlier detection within the context of IoT frameworks and Wireless Sensor Networks (WSNs), emphasizing the unreliability of sensor data and the potential of machine-learning techniques to overcome it. Within the realm of machine-learning-based outlier detection, two primary categories of techniques come to the forefront: statistical-based methods and classification-based methods. The authors suggest future directions that involve employing ensemble methods for outlier detection in IoT and WSN, utilizing machine-learning techniques and semantics to distinguish between errors and events, and integrating offline and online learning techniques to enhance anomaly detection.

The work in [19] proposes an online anomaly detection algorithm called *OnlineBPCA*, based on *Bilateral Principal Component Analysis (B-PCA)*, to address the challenge of real-time traffic anomaly detection in network management. They introduce several novel techniques within OnlineBPCA to support quick and accurate anomaly detection and suggest extensions of their approach to various IoT domains, including mobile and wireless sensor networks, social networks, and recommendation systems. In [20], deep-learning-based methods are leveraged to design an effective intrusion detection model for IoT networks. The authors employ *Convolutional Neural Network (CNN)* architectures for multiclass and binary traffic classification, enabling the detection of diverse attack types within IoT network traffic.

The study in [12] highlights the significance of *Multivariate Time Series Anomaly Detection (MTSAD)* in the context of the growing deployment of sensors in various environments. It reviews the current state of the art in unsupervised MTSAD, including advanced machine-learning and signal-processing techniques such as deep learning. The joint analysis of temporal patterns and measurement dependencies across different sensors—often exhibiting complex behavior—makes MTSAD a challenging problem. However, the authors note that many existing methods are tailored to specific use cases, and no single approach consistently generalizes across domains.

In [21], stream-based machine-learning models are applied to detect network attacks using real network traffic measurements from the WIDE backbone network, a high-capacity infrastructure that connects multiple smaller networks, such as Local Area Networks (LANs) and Wide Area Networks (WANs), across large geographical areas. Their findings indicate that certain models can effectively handle concept drift in real-time network data streams, improving both network security and anomaly detection capabilities.

The study in [22] further emphasizes that the continuous nature of IoT data and the need for online prediction, adaptation to concept drift, and early detection make traditional anomaly-detection methods unsuitable for real-time use. They advocate for continuous-learning algorithms that minimize false positives and negatives. Large-scale benchmarking is presented in [3], where 71 state-of-the-art anomaly

detection algorithms were evaluated across diverse domains using 976 time-series datasets. The findings indicate that, despite higher training costs, deep-learning models are not yet competitive in anomaly detection, performing comparably to or worse than simpler statistical approaches.

Complementary to these algorithmic perspectives, Li *et al.* [23] provide a detailed taxonomy of existing anomaly-detection methods, categorizing them according to their analytical principles. *Statistical approaches* distinguish normal from abnormal data by defining confidence boundaries, flagging values outside these limits as anomalies. Although simple and interpretable, they often fail to detect subtle or weak deviations. *Clustering-based algorithms*, such as k-means or DBSCAN, assume that normal data points form clusters, while points that do not belong to any cluster are considered anomalous. However, when abnormal data form their own cluster, these methods may incorrectly classify normal points as noise. *Density-based techniques*, including Outlier Detection using In-degree Number (ODIN), Relative Density-based Outlier Score (RDOS), and Local Outlier Factor (LOF), identify anomalies by comparing the local density of each data point with that of its neighbors. Despite their accuracy, their computational cost makes them less suitable for real-time engineering systems. *Ensemble methods*, such as Isolation Forest, are efficient and widely used due to their linear time complexity but mainly detect point anomalies and depend heavily on feature selection. In contrast, *distance-based approaches* are easier to implement and well-suited for time-series and signal-monitoring applications.

Within this category, one method in particular has gained considerable attention for its versatility and interpretability—the *Matrix Profile (MP)*. MP is a powerful data structure designed to measure similarity between subsequences in time-series data, making it highly effective for detecting anomalies in both univariate and multivariate datasets [4]. It enables efficient discovery of *motifs* (repeated patterns) and *discords* (unusual subsequences) within large data streams. Several studies have extended the Matrix Profile framework to enhance its performance and broaden its applications. For example, *Matrix Profile I* [4] introduced the concept through all-pairs similarity joins for time series, while *Matrix Profile II* [24] improved com-

putational efficiency using GPU acceleration. *Matrix Profile V* [13] incorporated domain knowledge into the matching process, and *Matrix Profile VII* [25] explored time-series chains to capture evolving patterns. *Matrix Profile XX* [26] expanded its capability to visualize motifs of varying lengths, and *Matrix Profile XXVIII* [27] extended the method to handle multidimensional data through a K-of-N anomaly detection strategy. Building on these advances, *Matrix Profile XXIV* [28] introduced DAMP, an algorithm that scales discord computation to ultra-large datasets, further demonstrating MP’s utility in time-series anomaly detection. A comprehensive overview of these developments is available in [29].

Further extending this line of research, Coulaud *et al.* [30] propose the *Interval Matrix Profile (IMP)* as an enhanced version of MP designed to analyze periodic and seasonal time series. Their work addresses a key limitation of the classical MP, which does not explicitly account for seasonality and may therefore miss recurrent seasonal anomalies. The IMP performs flexible interval-based comparisons across different seasonal cycles, enabling the detection of anomalies that occur periodically—an important feature for climate and environmental datasets. The authors also introduce a constrained k -Nearest Neighbor variant to identify anomalies appearing across multiple time periods, such as recurring extreme weather events. In addition, they develop a scalable, block-based implementation that leverages caching, vectorization, and parallelism to improve performance. Evaluated on climate temperature datasets, their method effectively captures seasonal anomalies and pattern disappearance, outperforming the standard MP in both accuracy and computational efficiency.

In summary, the literature on anomaly detection in IoT and time-series analysis demonstrates rapid methodological evolution—from rule-based and statistical models to data-driven and explainable frameworks such as the Matrix Profile. Recent advancements, including the Interval Matrix Profile, highlight a shift toward more adaptive and seasonality-aware techniques, which are particularly relevant for real-world IoT scenarios where data streams are dynamic, multidimensional, and often periodic.

Chapter 3

Methodology

3.1 Methodology

TBA

Chapter 4

Description of Datasets

4.1 Description of Datasets

TBA

Chapter 5

Experimental Evaluation

5.1 Experimental Evaluation

TBA

Chapter 6

Conclusions & Future Work

6.1 Conclusions

6.2 Future Work

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